

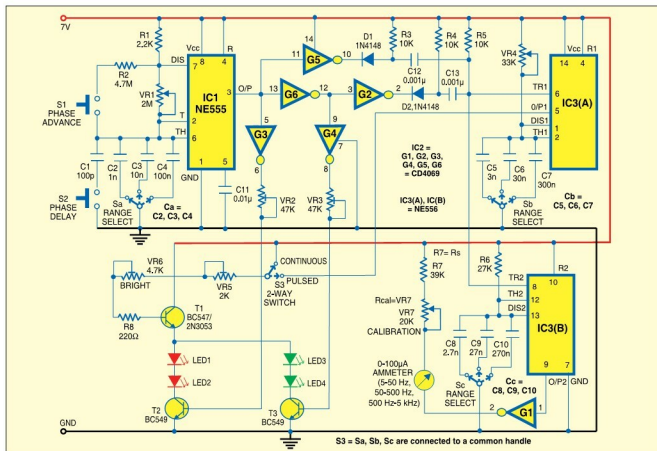


Fig. 1: Circuit of dual-colour stroboscope

Here is the circuit of a stroboscope that produces dual-coloured light pulses (refer Fig. 1). The circuit uses red and green LEDs as light sources to illuminate the object. You can choose the frequency of the stroboscope's light pulses from a wide range of 5 Hz to 5 kHz as desired. The range of frequency (5-50 Hz, 50-500 Hz, 500 Hz-5 kHz, etc) is selected through capacitors C_2 , C_3 and C_4 , C_5 , C_6 and C_7 and C_8 , C_9 and C_{10} .

Smooth variation of the frequency range is achieved by varying VR1. The length of the light pulses (both red and green pulses are of equal duration) is adjusted by VR4. A flash of red and green light makes one cycle of the stroboscope. The frequency meter reads this frequency.

The circuit comprises a free-running oscillator formed by IC NE555 (IC1), IC CD4069 (IC2), dual-timer IC NE556 (IC3) and a few discrete components. The pulse output at pin 3 of

NE555 triggers one-shot monostable IC3(A), which outputs a positive-going pulse that appears at the anodes of the LEDs (LED1 through LED4) through transistor T1.

However, gates G3 and G4 cause only one of the transistors T2 and T3 to turn on. So the positive-going pulse from monostable IC3 (A) causes either the red LEDs (LED1 and LED2) or the green LEDs (LED3 and LED4) to turn on, depending on the phase of the oscillator.

The astable multivibrator configured around IC1 is made to produce a symmetric square waveform by using the combination of low-value fixed resistor R1 (2.2-kilo-ohm) and high-value potentiometer VR1 (2-mega-ohm).

One-shot monostable IC3 (B) configured as a frequency meter triggers along with IC3(A). Strobe-phase delay or advance facility introduces a shift in the stationary orientation either clockwise or anti-clockwise when S2

is pressed momentarily. The 100µA ammeter is calibrated to read the frequency from 5 Hz to 5 kHz.

Presets VR5 and VR6 are used to set the maximum brightness of the LEDs. Potmeter VR7 is used to calibrate the frequency scale on the ammeter. The supply voltage should be regulated (7.0V) and there should be no voltage drop across the circuit when the LEDs glow. Any voltage drop affects the meter readings.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Fix the LEDs on the top or front side of the cabinet such that the LEDs' light falls on the moving object.

For testing, construct a paper disk with two white patches as shown in Fig. 2. Fit the disk onto a direct-current motor spindle.

Using switches S1 and S2, change the position of the patches in advance and delay motion, respectively. Pressing S1 slightly increases the frequency